W6 - ATT&CK Framework

Background & Threat Intelligence Context

To understand the practical mechanics of modern enterprise breaches, this lab focuses on emulating the tactics, techniques, and procedures (TTPs) associated with **Storm-0501**—a financially motivated threat group known for deploying ransomware and extortion campaigns against hybrid cloud environments.

Lab Environment & Network Setup

To simulate a realistic lateral movement attack scenario over WinRM (MITRE ATT&CK T1021.006) and fix the single-VM limitations from Week 5, a dual-VM network environment was configured in VirtualBox:

- **Attacker System (Ubuntu Linux VM):** Represents the threat actor's machine hosting the post-exploitation framework Evil-WinRM.
- **Victim System (Windows 11 VM - WinDev2407Eval):** Represents the compromised internal target host with the WinRM service enabled, Sysmon telemetry active, and Splunk Enterprise collecting logs.

Step 1: Network Reachability & Service Verification

First, we verified that the attacker VM (Ubuntu) had direct network visibility and connectivity to the target victim machine (192.168.0.6) over the standard unencrypted WinRM port (5985).

1. **Target IP Discovery (Victim):** Executed ipconfig on the Windows target VM to confirm its assigned network IP address (192.168.0.)

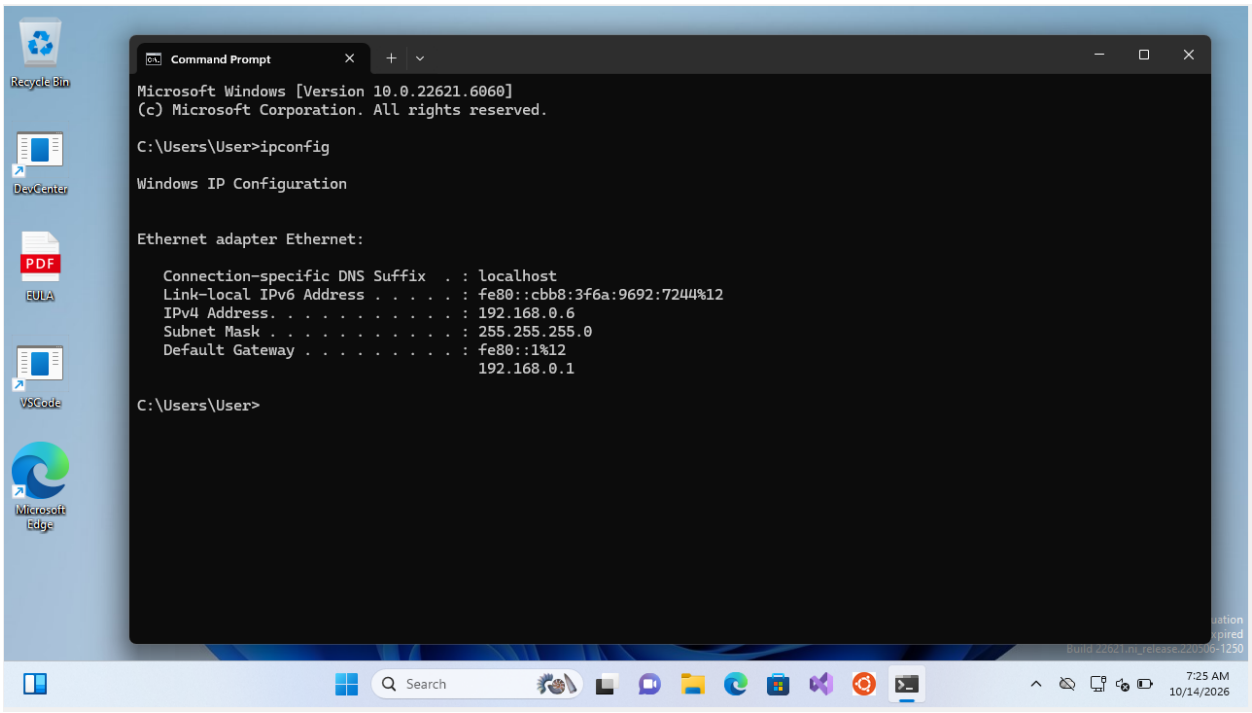


Figure 1: *Target Windows VM (WinDev2407Eval) IP configuration showing 192.168.0.6*

2. **Network Ping & Port Scanning (Attacker):** From the Ubuntu terminal, we performed an ICMP ping check to confirm layer-3 reachability:

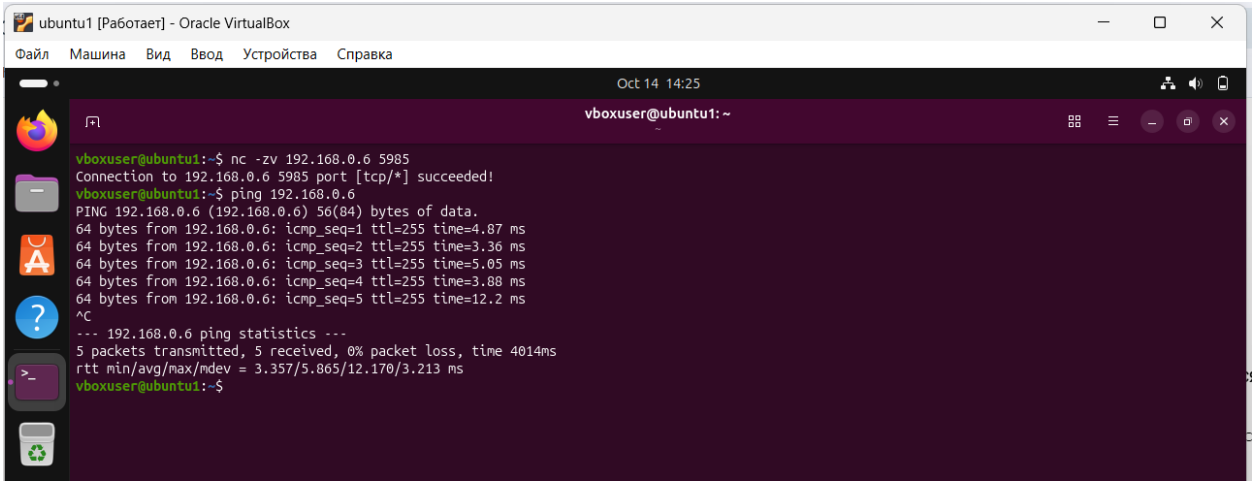


Figure 2: *Successful network reachability (ping) and open WinRM port 5985 check (Netcat) from the Ubuntu attacker VM.*

Step 2: Establishing a Remote Session via Evil-WinRM

To emulate the lateral movement behavior of the Storm-0501 ransomware group (MITRE ATT&CK T1021.006) from an external attacker node rather than locally, we used the evil-winrm post-exploitation framework installed on the Ubuntu VM.

1. **Tool Execution:** Executed the following command from the Ubuntu attacker node against the target Windows victim IP using valid administrative credentials:

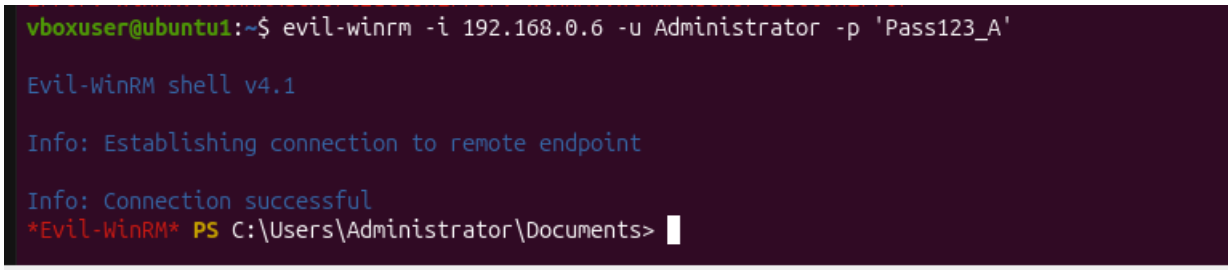


Figure 3: *Successful installation of Evil-WinRM on the Ubuntu attacker node and initiation of the remote session connection to the Windows victim VM (192.168.0.6)*

Step 3: Execution of Storm-0501 Discovery Commands

After establishing the remote session via Evil-WinRM from the Ubuntu attacker machine (192.168.0.x) to the Windows target (192.168.0.6), a suite of reconnaissance commands was executed:

1. **Privilege & Service Discovery:**
   - whoami -Confirmed the active context as windev2407eval\administrator

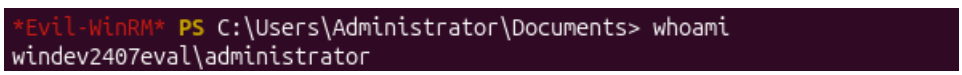


Figure 4.1: *Verification of active administrative privilege context via whoami inside Evil-WinRM.*

- sc query sense & sc query windefend -Checked the status of Microsoft Defender Antivirus and Defender for Endpoint services to evaluate host defenses.

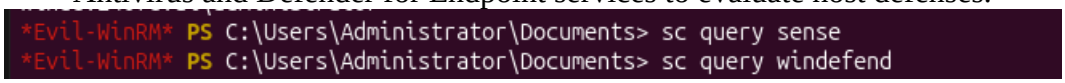


Figure 4.2: *Querying security service status (sense and windefend) to identify active endpoint detection controls.*

2. **Account & Network Reconnaissance:**
   - net user & net localgroup administrators -Enumerated local user accounts and identified accounts with administrative privileges (Administrator, User).

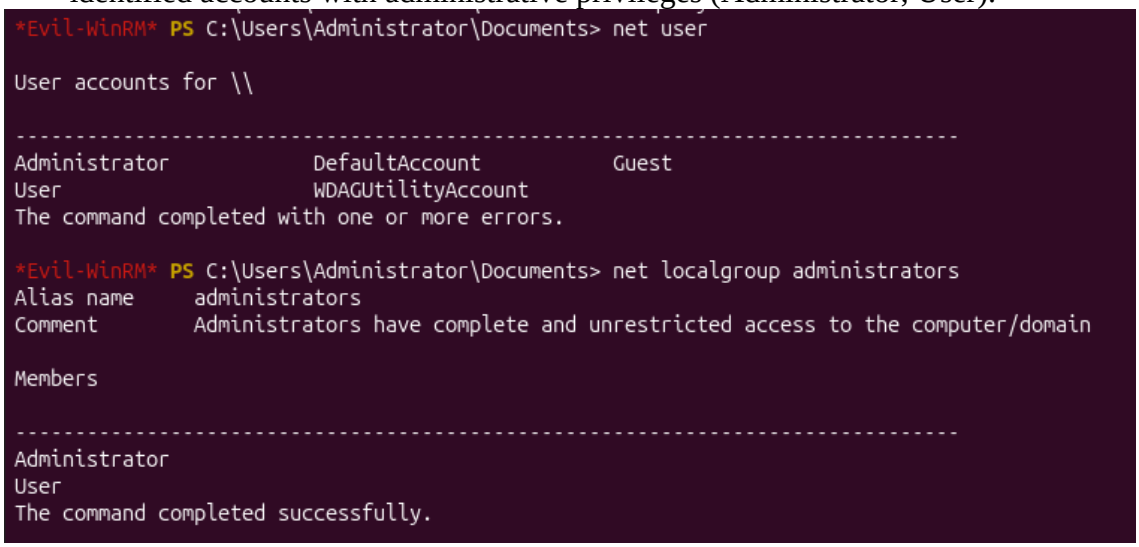


Figure 4.3: *Enumeration of local user accounts and administrative group memberships.*

- ipconfig /all -Gathered network interface settings, DNS servers, and IP routing configuration.

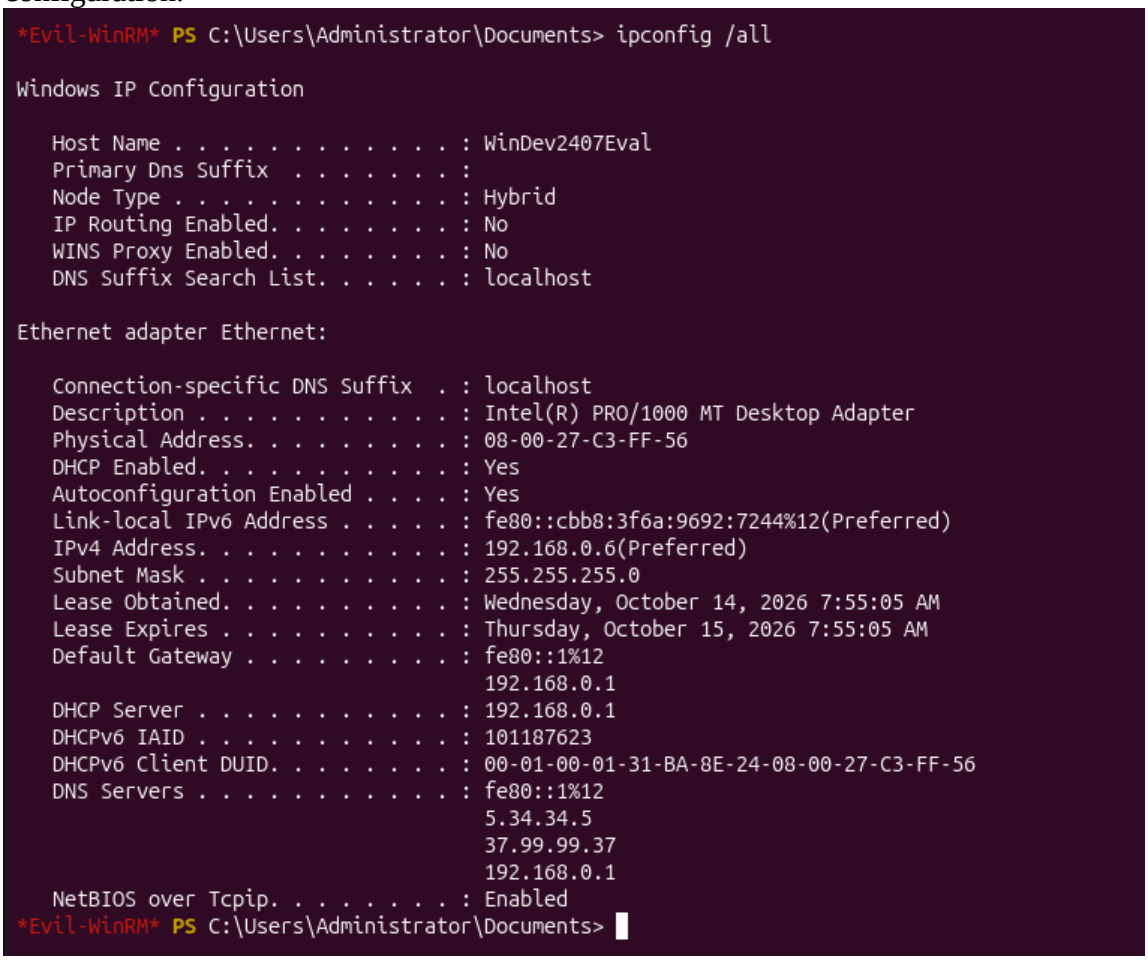


Figure 4.4: *Host network adapter and routing discovery executed over remote WinRM channel.*

Detection: Evidence Collected in Splunk

Unlike Week 5's localhost simulation, this setup represents a true multi-host network attack over port 5985. Telemetry on the victim host now records an external network connection (Sysmon EID 3) paired with remote process execution under wsmprovhost.exe (Sysmon EID 1)

Time	Source IP	Destination IP	Dest. Port
08:15:07.013	192.168.0.4 (Ubuntu attacker)	192.168.0.6 (Windows victim)	5985
08:15:07.252	192.168.0.4	192.168.0.6	5985

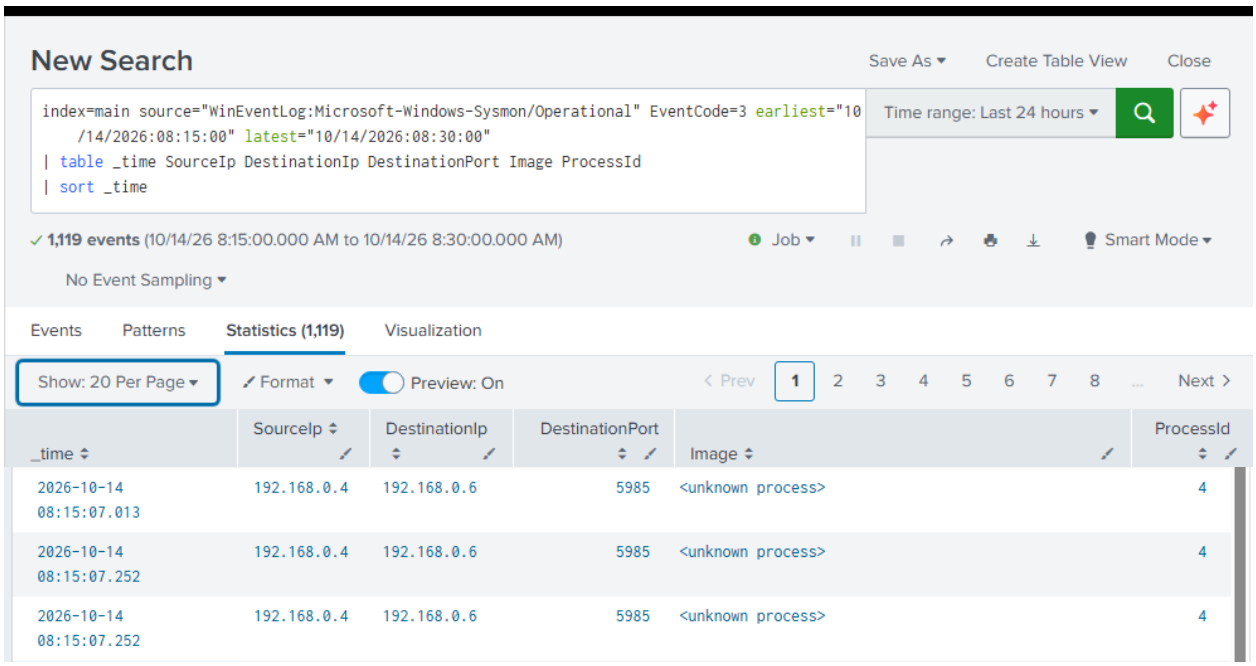


Figure 5. Sysmon EID 3 showing the network connection from the attacker VM (192.168.0.4) to the victim VM (192.168.0.6) on port 5985.

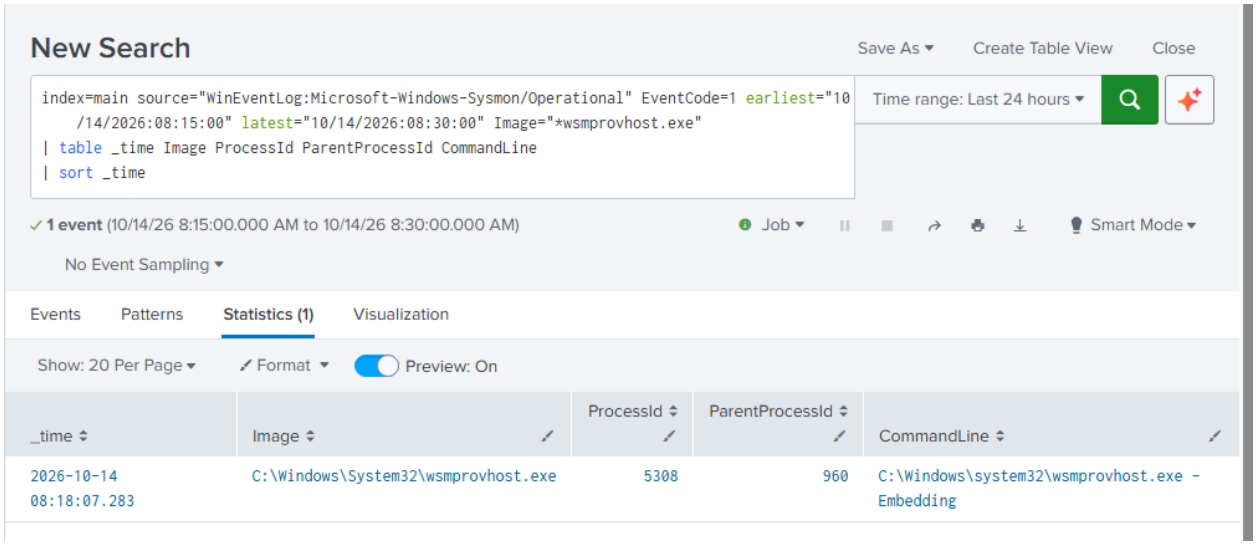


Figure 6. Find the wsmprovhost.exe process itself and the entire session duration

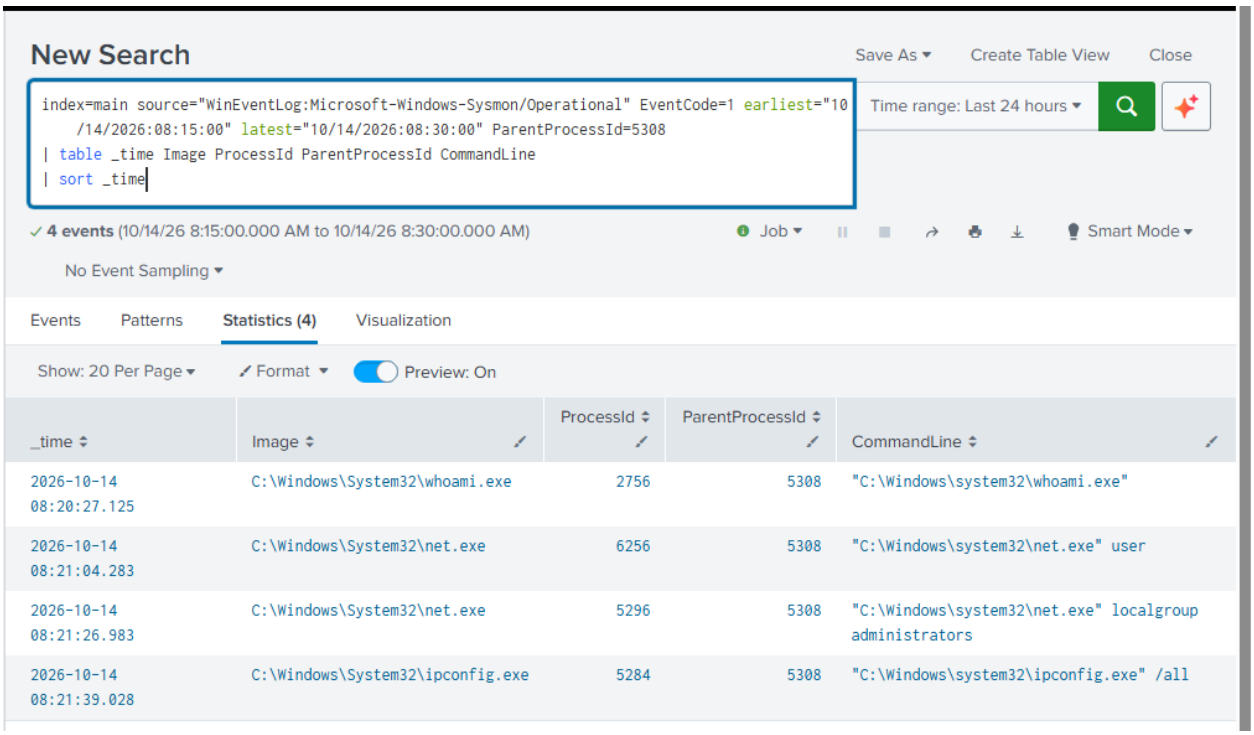


Figure 7 Children of wsmprovhost.exe (PID 5308 was already seen, but let's check for any potential new ones as well)

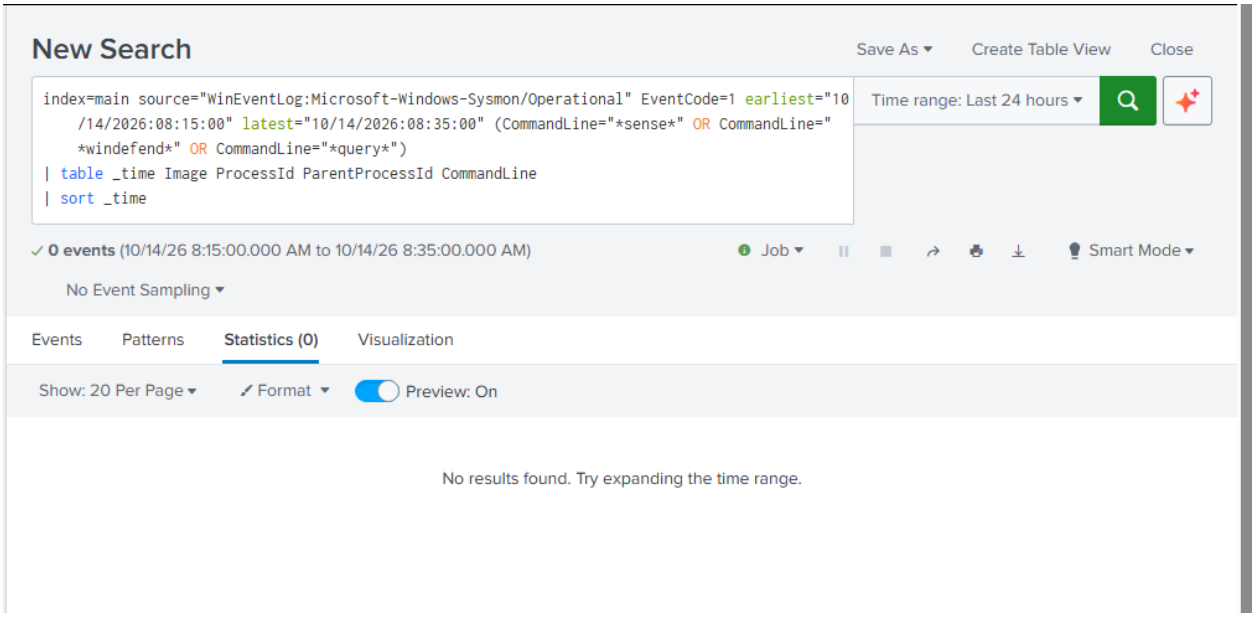


Figure 8 sense, windefend, and query are also empty -this means it's a solid finding: the sc query command in the Evil-WinRM session did not leave a separate process in Sysmon.

Mapping to ATT&CK Tactics and Techniques

Each action performed in Steps 2–3 is mapped below to its MITRE ATT&CK tactic and technique, together with the Splunk evidence (or lack thereof) that supports it.

Action	Evidence	Tactic	Technique
Network connection, Ubuntu → Windows:5985	Sysmon EID 3, 08:15:07, 192.168.0.4→192.168.0.6:5985	Lateral Movement	T1021.006
Evil-WinRM session established	Sysmon EID 1, wsmprovhost.exe PID 5308, parent svchost.exe (992→960 after VM reboot), 08:18:07	Lateral Movement / Execution	T1021.006 / T1059.001
whoami	Sysmon EID 1, parent PID 5308, 08:20:27	Discovery	T1033 (System Owner/User Discovery)
net user	Sysmon EID 1, parent PID 5308, 08:21:04	Discovery	T1087.001 (Local Account)
net localgroup administrators	Sysmon EID 1, parent PID 5308, 08:21:26	Discovery	T1069.001 (Local Groups)
ipconfig /all	Sysmon EID 1, parent PID 5308, 08:21:39	Discovery	T1016 (System Network Config. Discovery)
sc query sense / sc query windefend	Executed per Evil-WinRM output (Fig. 4.2); no corresponding process in Sysmon EID 1	Discovery	T1518.001 (Security Software Discovery) -not confirmed in telemetry

This set of commands overlaps closely with the discovery activity Microsoft documents for Storm-0501 in MS-2025 (systeminfo, net, nltest, tasklist, and queries of Defender services -see Week 4, Section 3, stage 4/6), while also including two techniques (T1033, T1069.001) not explicitly listed in the Week 4 technique table. This is discussed as a partial-overlap finding in Section 6.

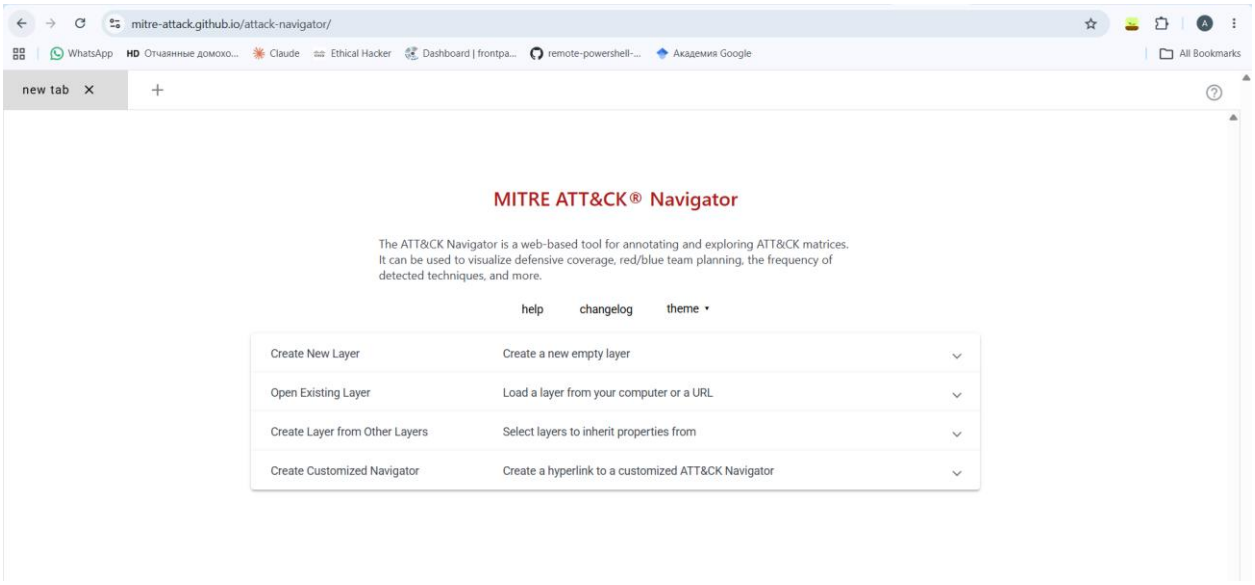
ATT&CK Navigator: Documented vs. Observed

To visualize how the lab's results compare against the full set of techniques MITRE documents for Storm-0501 (group page G1053, as compiled in the Week 4 report), two Navigator layers were built and overlaid:

- Layer A -Documented: all techniques listed for Storm-0501 in the Week 4 Kill Chain table (28 techniques across MS-2024 and MS-2025), sourced from MITRE ATT&CK G1053.

- Layer B -Observed: the techniques reproduced and confirmed in this lab via Sysmon (Section 3).

The overlay uses four colors: green -documented for Storm-0501 and confirmed in this lab's telemetry; purple -observed in this lab but not present in the Week 4 technique list; blue -documented for Storm-0501 but not reproduced in this lab (e.g., cloud pivoting, DCSync, Rclone exfiltration, ransomware deployment); yellow -attempted in the lab (per terminal output) but not confirmed in Sysmon telemetry.



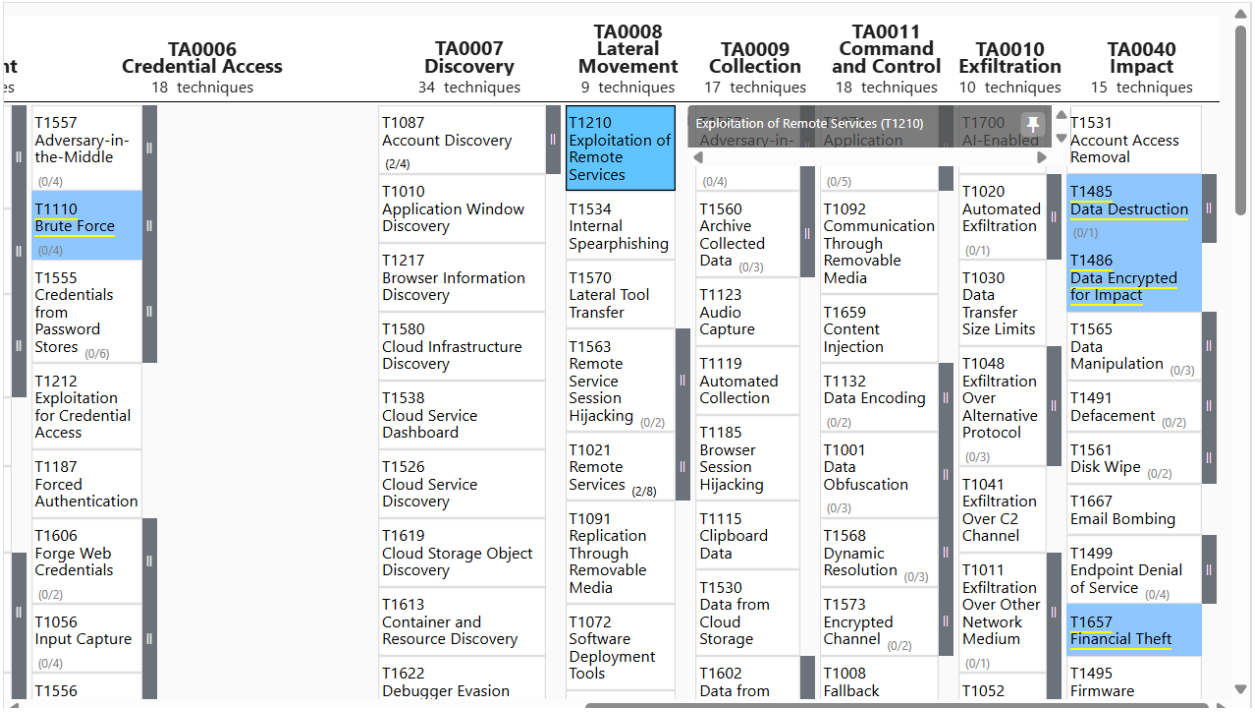


Figure 9. ATT&CK Navigator overlay. T1021.006 and T1059.001 (Lateral Movement / Execution) are marked green, confirmed in both Storm-0501 reporting and lab telemetry. T1033

and T1016 (Discovery) are marked purple. T1485 Data Destruction and other Actions-on-Objectives techniques are marked blue, documented for the group but out of scope for this lab.

Conclusion

This lab emulated the lateral-movement and discovery stage of the Storm-0501 attack chain (Kill Chain stage 6, Week 4) in a dual-VM environment, using Evil-WinRM from an external attacker node to execute Storm-0501-style discovery commands against a Windows victim over WinRM (T1021.006). Two independent Sysmon event types -EID 3 (network connection) and EID 1 (process creation, with wsmprovhost.exe as parent) -together provided end-to-end evidence of the attack, from the initial network connection through to each discovery command. Four of the seven mapped actions align directly with techniques Microsoft documents for Storm-0501; the remainder extend the Week 4 technique list or, in the case of sc query, could not be confirmed in telemetry at all despite confirmed execution. The ATT&CK Navigator overlay (Section 5) makes this coverage, and its limits, visible at a glance: the lab closes part of the Lateral Movement and Discovery picture for Storm-0501 while leaving the group's broader cloud-persistence and impact techniques unaddressed.